

Yao-An Yang's Statement of Purpose

Since my introduction to physics, I have always been intrigued by how complex patterns arise from simple principles. This curiosity now leads me to my interest in Machine Learning, where models learn meaning and structure from limited data. My research interest spans **representation learning**, **multimodality**, and **mechanistic understanding**. In the long term, I aim to build and understand **intelligent systems that can continually and efficiently learn with natural supervision and embodied interactions**. I approach this complicated problem in both a scientific and engineering viewpoint, where I view these two principles as mutually reinforcing processes. While scientific testing provides insights into how models learn and how we should design them, engineering provides more capable models and new phenomena to analyze and explain.

Scientifically, I am curious about how properties of data, such as structure and modality, interact with model architecture and training to shape **representation learning**. Developing such understanding requires examining models not only in terms of aggregated performance, but through localized analysis and how their internal representations change during training and inference. Inspired by this philosophy, I began my research journey working with Prof. Danai Koutra, where we studied how distribution of node features around specific nodes, which we termed feature heterophily, affected link prediction performances of Graph Neural Networks. I contributed to the implementation of various models and experiments, and helped identify effective design choices under specific data properties. The work led to a conference publication at NeurIPS 2024.

In terms of model building, I believe that scaling law will allow us to build stronger models. However, it is crucial to study how we can build more **data-efficient models** that scale well with more computation. Combined with the recent advent of applying Large Language Models for graph learning on text-attributed graphs, I set on to explore efficient ways to fuse LLMs and GNNs under the guidance of Prof. Koutra. We discovered that while GNNs are affected by their inductive biases in graph structure, LLM's performance remained largely constant across the whole graph. This inspired us to design a model that learns to use the more efficient GNN in most regions, and only apply LLMs to nodes where GNNs are predicted to perform poorly. This not only improved the performance of the model by 13% on heterophilous nodes, but also allowed us to apply our algorithm to graphs as large as 2.5M nodes, which is computationally infeasible for previous LLM-aided graph learning pipelines. The work is currently under review for ICLR 2026.

Additionally, I am interested in exploring **embodied or multimodal contexts** as sources of richer learning signals. This has led me to work with Prof. Joyce Chai where we are currently exploring an efficient model steering method to enable Vision-Language Models to learn visual and textual correspondences. Specifically, we implemented a lightweight transformer head on top of a frozen VLM and trained it on the Localized Narrative dataset. By training on this dense correspondence between points on images and textual descriptions, we aim to improve VLM's attention to detail and allow it to generate conditioned captions based on different locations on the image.

Working with multimodality, I observed that while many recent works have shown that models exhibit traces of learning emergent meaning across modalities, few have rigorously tested whether the correlations they learn reflect genuine abstraction or merely statistical association. This observation sparked my interest in a **mechanistic view of learning**—examining whether and how their internal representations track state, form abstract structures, or evolve with data properties. This interest has been further deepened in the interdisciplinary project I worked on in collaboration with Prof. Joyce Chai and Prof. Jonathan Brennan in the Neurolinguistics department. We investigated claims that drew comparisons between human's working memory and LLM's in-context abilities to track internal state of beliefs. For LLMs, we started by prompting the model with various patterned text inputs and checked their probability distribution and discovered that some models learned the pattern quickly after a few repetitions while others struggled. Interestingly, we also saw some models exhibiting approximate counting, where they predicted a change in pattern but not in accurate positions.

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To explain these findings and variations across models, we are currently working on training smaller LMs from scratch and studying changes in internal representations during training.

All my experiences and curiosity have helped create a clearer picture of my research interests, and prepared me for pursuing graduate studies. In my graduate studies, I wish to continue working in the intersection of **representation learning**, **multimodality**, and **mechanistic interpretability** to push the boundaries of understanding intelligence and build models that can learn from the rich multimodal world. While most current multimodal models are limited to processing and reasoning primarily in one modality and using the other modality as auxiliary information, I aim to enable models to truly **utilize the best of both modalities** and seamlessly reason across modalities to gain the most information. I believe this will require first understanding whether and how the models have robustly learned abstract concepts across modalities by examining the evolution of internal representations during training, and then creating models that can uniformly understand and generate multimodal representations. More concretely, here are the research questions I wish to address: (i) How do we utilize multimodal data to enable more efficient learning? (ii) What data properties and model designs enable robust multimodal learning and reasoning? (iii) How can we probe models to ensure the correlations learnt are robust and how can we understand models beyond benchmarking?

I see the pursuit of graduate studies as both a challenge and an opportunity to refine my theoretical understanding, expand my research perspective, and deepen my ability to connect analysis with design. I aspire to not only advance model performances, but to also uncover the mechanisms that make models learn efficiently. Through this pursuit, I hope to contribute to a more coherent understanding of how intelligent systems form, represent, and update knowledge, bridging the gap between the science and the engineering of learning.